

Chemistry — Assignment 9: Electronic Effects

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Question. Differentiate inductive effect, resonance/mesomeric effect, electromeric effect and hyperconjugation.

Effect	Nature	Description
Inductive	permanent	displacement of σ -bond electrons along a chain due to electronegativity difference; weakens with distance (+I / -I)
Resonance / Mesomeric	permanent	delocalisation of π electrons or lone pairs in a conjugated system (+M / -M)
Electromeric	temporary	complete transfer of a π -electron pair to one atom when a reagent attacks; reversible (+E / -E)
Hyperconjugation	permanent	delocalisation of $\sigma(\text{C-H})$ electrons into an adjacent empty orbital / π bond ("no-bond resonance"); more $\alpha\text{-H} \rightarrow$ more stable

Prepared by Anshuman Atrey · structures rendered with RDKit.